

CMPE-255 Data Mining Project Report DRAFT

Title - Mental Health & Crisis Signal Mining on Social Media

Multi-class classification and interpretability analysis on social media mental-health statements

GitHub Link: <https://github.com/NateyLB/CMPE255Project>

1. Project Team

Course	CMPE 255 – Section 34, Spring 2026
Team Members	Ravikumar Komandur Narayanan – SJSU ID: 018265117 Nathan Howland – SJSU ID: 014826357 Gunanidhi Ramakrishnan – SJSU ID: 019108284 Jesigga Sigurdardottir – SJSU ID: 019156826

2. Dataset

A publicly available dataset from Kaggle is used in this project. It is freely accessible under permissive licenses, requires no IRB or data-use agreement, and is loaded directly via the Kaggle API using pandas. The dataset is cleaned and preprocessed, then split into a training and test dataset.

Dataset URLs:

Sentiment Analysis for Mental Health (Sarkar):

<https://www.kaggle.com/datasets/suchintikasarkar/sentiment-analysis-for-mental-health>

Dataset	Size	Labels	Role in Project
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Sentiment Analysis for Mental Health (Sarkar)	~51,000 statements	Seven classes: Normal, Depression, Suicidal, Anxiety, Stress, Bipolar, Personality Disorder	Used for training and in-distribution test set (stratified 80/20 split) All 7 classes were used for training and evaluation by a stratified 80/20 split (42,144 train / 10,537 test).
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3. Abstract - Project Idea

Mental health conditions, including depression, anxiety, bipolar disorder, personality disorder, and chronic stress, affect hundreds of millions of people worldwide, yet remain significantly underdiagnosed because of stigma, cost, and limited access to care. Social media platforms, such as Reddit, have become spaces where individuals openly express symptoms and crises, producing large volumes of self-disclosed text suitable for automated analysis. This project investigates whether classical and transformer-based natural language processing techniques can reliably classify media posts into mental-health conditions and how those models behave.

We built a multi-class classification pipeline trained on labeled posts from the Sarkar Sentiment Analysis Dataset, with seven labels: Stress, Depression, Bipolar Disorder, Personality Disorder, Suicidal, Normal, and Anxiety. We compare three classical machine-learning approaches (Logistic Regression, Linear SVM, and XGBoost over TF-IDF and engineered features) against a fine-tuned DistilBERT transformer to quantify the gain from contextual representations. Latent Dirichlet Allocation is applied for interpretable topic discovery, and SHAP token-level attributions provide model explainability for the best-performing classifier. A lightweight Streamlit application provides a live inference interface for demonstration.

4. Introduction

4.1 Motivation

The global burden of mental illness is staggering. The World Health Organization estimates that one in eight people worldwide lives with a mental health disorder, yet the majority never receive a diagnosis or treatment. Stigma, cost, and geographic barriers to care mean that many individuals only seek help after a crisis has already escalated.

Social media platforms such as Reddit have become important informal support channels. Users post candidly about depressive episodes, anxiety, bipolar swings, personality-disorder traits, and chronic stress in dedicated subreddits. This produces a large, naturally labeled dataset of self-disclosed mental-health text suitable for automated screening research. At the same time, deploying any such classifier in a setting that affects real people raises immediate questions about generalization across data sources and about behavior on out-of-distribution inputs.

4.2 Objective

This project has three core objectives:

- Build a robust 7-class NLP classification pipeline that distinguishes Stress, Depression, Bipolar Disorder, Personality Disorder, Suicidal, Anxiety, and Normal with a high macro-averaged F1 score on a stratified in-distribution test set.
- Compare classical bag-of-words models (TF-IDF + Logistic Regression, Linear SVM, and XGBoost) against a fine-tuned contextual transformer (DistilBERT) to provide an empirical analysis of the accuracy / precision-recall / computational-cost trade-off.

Throughout, model interpretability is provided via SHAP token-level attributions and LDA topic visualizations, supporting the auditability of model decisions.

4.3 Technical Challenges

Several challenges shape the design of this pipeline:

- **Class Overlap and Lexical Confusion.** Stress, Depression, and Anxiety language overlap considerably (vocabulary of fatigue, worry, hopelessness, and rumination is shared across all three). Bipolar Disorder and Personality Disorder posts tend to use more clinical vocabulary phrasing, but still share emotional registers with the other classes. Fine-grained multi-way separation is therefore harder than a coarse condition-vs-control task.
- **Class Imbalance.** The classes are not balanced, so we decided to undersample the dataset to create a balanced training dataset.
- **Distributional Shift Across Sources.** Models trained on subreddit-derived labels may not generalize to Media Text statements, which use a different annotation methodology and a different statement-level distribution. This is why we are using a combined dataset to create a more robust model.
- **Interpretability vs. Accuracy Trade-off.** DistilBERT is expected to outperform classical baselines on raw accuracy but is harder to explain. SHAP token-level attributions and LDA topic clusters bridge this gap and produce auditable evidence behind individual predictions.

5. Problem statement

Given the Media text dataset, the primary task is to assign it to one of $K = 7$ classes:

- C_1 = Stress
- C_2 = Depression
- C_3 = Bipolar Disorder
- C_4 = Personality Disorder
- C_5 = Anxiety
- C_6 = Normal
- C_7 = Suicidal

Formally, we seek a classifier $f: X \rightarrow \{C_1, C_2, C_3, C_4, C_5, C_6, C_7\}$ where X is the space of raw post texts, trained to maximize macro-averaged F1 over a stratified 80/20 train-test split of the dataset. Macro-averaging is used deliberately rather than accuracy or micro-F1 so that minority-class performance is not masked by the larger classes.

Auxiliary goals: discover the latent topical structure of each class via LDA and explain individual classifications via SHAP token-level attributions.

6. Description of Steps

Step 1 – Data Acquisition & EDA

The EDA notebook begins by loading the social media-style datasets into pandas DataFrames. The primary goal at this stage is to understand the structure of the data, including the number of samples, available features, and overall schema consistency. Basic inspection steps, such as viewing dataset shapes, column names, and sample rows, are performed to establish an initial understanding of the data. We then analyze label distribution across mental health categories. This step highlights a significant class imbalance, where some classes (such as normal) dominate while crisis-related categories are underrepresented. This imbalance is visualized using plots or bar charts, which helps later in preprocessing and modeling decisions, such as class weighting or resampling. Next, the EDA focuses on text quality. The dataset is checked for missing or null values in the text fields, as well as extremely short or unusually long posts. These checks help identify noisy or incomplete data entries that may impact model training. The variability in post length is also examined to understand the natural distribution of social media text. Overall, the EDA identifies key issues such as class imbalance, noisy text, and heterogeneous data sources, which directly inform the preprocessing strategy.

Step 2 – Data Preprocessing

The preprocessing notebook loads the Sarkar dataset, standardizes column names and label structures into a uniform format, and applies a stratified 80/20 train-test split. This step ensures that the dataset is consistently formatted and ready for model training. Text cleaning is then applied to all posts. This includes converting text to lowercase, removing URLs, mentions, hashtags, punctuation, and other special characters. The goal is to reduce noise found in social media text data while preserving the meaning of the text. Extra whitespace is also normalized to ensure consistent formatting. Then, cleaning, tokenization, and text normalization are performed. Finally, the dataset is prepared for modeling by shuffling and splitting into training and test sets. The preprocessing outputs a clean dataset containing processed text and encoded labels. This dataset is then ready for feature extraction and model training.

Step 3 – Feature Engineering

Four families of features are constructed from the cleaned training corpus:

- **TF-IDF Vectorization.** Unigrams and bigrams, `max_features = 50,000`, `sublinear_tf = True`. The vectorizer is fit on the training split only and persisted to `artifacts/tfidf_vectorizer.pkl`; train and test transforms are saved as sparse matrices.
- **LIWC-Inspired Lexicon Counts.** Per-post counts of words from public emotional-, cognitive-, and social-process category lexicons (NRC Emotion Lexicon or equivalent freely licensed list).
- **Post-Level Surface Statistics.** Word count, sentence count, punctuation density, capitalization ratio, and average word length serve as auxiliary signals of emotional intensity.
- **LDA Topic Probabilities.** After Step 5, the per-post topic probability vector from the optimal LDA model is appended as supplementary features for the EXP-3 ablation.

Step 4 – Model Training & Comparison

Four models are trained on identical train splits and evaluated on the same test set:

- Logistic Regression with TF-IDF features, baseline. C tuned by 5-fold stratified cross-validation over `{0.01, 0.1, 1, 10}`, L2 penalty.
- Linear SVM with TF-IDF features. C tuned by 5-fold stratified cross-validation.
- XGBoost over TF-IDF concatenated with engineered features (LIWC + post-level + LDA topic probabilities). Hyperparameters tuned with Bayesian search (Optuna or BayesSearchCV) over `max_depth`, `learning_rate`, `n_estimators`, `subsample`, `colsample_bytree`, and `min_child_weight`.
- DistilBERT (`distilbert-base-uncased`) fine-tuned for up to 3 epochs with AdamW (`lr = 2e-5`, `batch size = 32`, `weight decay = 0.01`, `warmup ratio = 0.1`) using the HuggingFace Transformers Trainer API. A 10% validation slice is carved from the training split for early stopping on validation macro-F1; the reported number is the best epoch's metric, not necessarily the third.

Each classical model is run under three balancing conditions (no balancing, class-weighted, and SMOTE), and metrics for every condition are appended to a shared results/metrics.csv with a fixed schema. A separate stylistic-features-only leakage probe (Logistic Regression on post-level surface statistics with no lexical content) is run as a sanity check against subreddit-style leakage; macro-F1 from this probe is reported alongside the lexical models.

Step 5 – Topic Modeling

Latent Dirichlet Allocation is fit with $k \in \{8, 10, 12, 15, 18, 20\}$ topics. The optimal k is selected by maximizing the C_v coherence score. The fitted model is then used to (a) generate per-post topic probability vectors that feed back into Step 3 for the XGBoost feature ablation, (b) cross-tabulate topic-by-class to identify which latent themes (e.g., isolation, hopelessness, manic energy, identity disturbance, somatic anxiety) co-occur with which trained label, and (c) produce a pyLDAvis HTML visualization for inclusion in the final paper.

Step 6 — Evaluation, Cross-Source Probe, OOD Safety, and Explainability

Each model is evaluated on the test set using macro-averaged F1, weighted F1, per-class precision and recall, ROC-AUC (one-vs-rest, macro-averaged), and a 7×7 confusion matrix. The best macro-F1 model from EXP-1 through EXP-6 is then carried forward.

SHAP TreeExplainer is applied to the XGBoost model to produce global feature importance and force plots for representative correctly classified and misclassified examples per class. The SHAP partition explainer is applied to the DistilBERT model to produce token-level attribution heatmaps over the same examples.

A lightweight Streamlit application provides live inference: a user pastes text, sees the predicted class, the per-class confidence vector, and the top SHAP-attributed tokens. A persistent disclaimer on every page makes clear that the demo is for academic demonstration only and is not intended for clinical screening.

7. Related work

7.1 Classical NLP Approaches to Mental Health Detection

Coppersmith et al. [1] established an influential baseline through the CLPsych 2015 shared task on Twitter, demonstrating that TF-IDF-weighted n -gram features combined with standard linear classifiers (SVM, Logistic Regression) can reliably distinguish depression and PTSD posts from controls. Their experimental configuration informs our classical baseline setup (EXP-1 and EXP-2).

7.2 Subreddit-Style Leakage and Weak Supervision

Low et al. [3] document the risk that classifiers trained on subreddit-derived labels can achieve very high accuracy by learning surface stylistic cues (moderation templates, trigger-warning prefixes, in-group vocabulary) rather than genuine mental-health signals. This motivates the explicit preprocessing-stripping step in our Step 1, the per-class precision and recall reporting alongside macro-F1, and the stylistic-features-only leakage probe described in Step 4.

7.3 Deep Learning and Transformer Models

Ji et al. [2] survey machine-learning methods for suicidal ideation and adjacent mental health detection and report consistent macro-F1 gains of 5 to 15 percentage points for fine-tuned BERT-family models over classical baselines. Sanh et al. [4] introduce DistilBERT, a 40% smaller and 60% faster distillation of BERT-base that retains roughly 97% of BERT's downstream performance — making it a practical transformer choice for a course project with constrained GPU budget. These results motivate EXP-4 in our experimental design.

7.4 Model Interpretability and SHAP

Lundberg and Lee [5] introduce SHAP, a unified game-theoretic framework for additive feature attribution that subsumes earlier methods (LIME, DeepLIFT) under a single Shapley-value formulation. SHAP has become a de facto standard for instance-level interpretability and has been applied to mental-health NLP to surface lexical drivers of model predictions [2]. We adopt SHAP TreeExplainer for XGBoost and SHAP partition explainer for DistilBERT, supporting auditability of individual classifications.

8. Solution

Our solution is an end-to-end NLP pipeline with six stages, mapped to the steps in section 6 and the experiments in section 9:

- **Data integration and cleaning.** Data integration and cleaning. The Sarkar Sentiment Analysis dataset is loaded, cleaned (lowercasing, URL/punctuation removal, whitespace normalization), and split into stratified 80/20 train/test sets preserving the 7-class label distribution.
- **Feature representation.** Dual-track features: TF-IDF + LIWC + surface statistics (and LDA topic probabilities, added in the EXP-3 ablation) for classical models; raw tokenized sequences for DistilBERT fine-tuning.
- **Multi-model comparison.** Four models trained under identical conditions on the same train-test split, enabling a controlled ablation: LR (baseline) → SVM → XGBoost → DistilBERT (transformer).
- **Topic modeling.** LDA with coherence-based k selection provides class-level interpretable topic clusters as a supplementary analytical layer and as input features for the XGBoost ablation.

- **Explainability and deployment.** SHAP attributions on the best in-distribution model and on test data; Streamlit app for live demonstration with a clear non-clinical disclaimer.

The pipeline is implemented in Python using pandas, scikit-learn, imbalanced-learn, XGBoost, gensim, pyLDAvis, HuggingFace Transformers, PyTorch, SHAP, and Streamlit. All notebooks, fitted artifacts, and a shared metrics CSV are version-controlled in the project GitHub repository.

9. List of experiments

All experiments use a single shared experimental protocol:

- Train/Test Split: Stratified 80/20 split on the media text dataset, persisted to parquet, and read identically by every notebook.
- Cross-Validation: 7-fold stratified CV on the training set for hyperparameter tuning.
- Class Imbalance: No-balancing, class_weight="balanced", and SMOTE on the training split only, all three reported.
- Feature Ablation: TF-IDF only versus TF-IDF + engineered features (within EXP-3) to isolate the contribution of hand-crafted features.
- Random Seed: A single RANDOM_SEED = 42 is used in every notebook (numpy, random, torch, transformers) for reproducibility.

Experiment	Model / Method	Feature Input	Macro-F1 (7-class)
EXP-1 (Baseline)	Logistic Regression	TF-IDF (unigrams + bigrams)	0.7502
EXP-2	Linear SVM	TF-IDF (unigrams + bigrams)	0.7520
EXP-3	XGBoost (Bayesian-tuned)	TF-IDF + engineered (LIWC, surface, LDA)	0.7965
EXP-4	DistilBERT (fine-tuned)	Raw token sequences	Pending

10. Proposed Evaluation Metrics and Justification

Metric	Justification
Macro-averaged F1	Treats each of the seven trained classes equally, regardless of support. Primary ranking metric for selecting the best model.
Per-class Precision / Recall	Precision controls false alarms; recall controls missed cases. Both are clinically meaningful for the three lexically overlapping classes (Stress, Depression, Anxiety) and for the two more clinical classes (Bipolar Disorder, Personality Disorder).
ROC-AUC (per class, one-vs-rest, macro-averaged)	Measures separability between each class and the rest, providing a threshold-independent ranking metric to complement macro-F1.
Confusion Matrix	Provides a full 7×7 error breakdown showing which classes are most often confused. Informs feature engineering and preprocessing follow-ups.
Cross-source Macro-F1 Gap (EXP-5)	The drop from in-distribution macro-F1 to cross-source macro-F1 on the matched Media Text labels is a direct measure of distributional robustness.
LDA Coherence (C_v)	Selects the optimal number of latent topics k by measuring the semantic coherence of the top words per topic.
SHAP Token Importance	Provides instance-level interpretability by attributing predictions to individual tokens — essential for auditability in any deployment-adjacent context.

Primary ranking metric: macro-averaged F1 on the in-distribution test set. Secondary headline: cross-source macro-F1 gap (EXP-5).

11. Results

11.1 Exploratory Data Analysis – Initial Findings

The dataset has been loaded and explored. The full EDA notebook is available in the GitHub repository at [notebooks/01_EDA.ipynb](#).

Dataset sizes:

- Sentiment Analysis for Mental Health (Sarkar): ~51,000 statements across seven classes (Normal, Depression, Suicidal, Anxiety, Stress, Bi-Polar, Personality Disorder).

Key initial observations:

- Bipolar Disorder and Personality Disorder posts have visibly longer average post lengths and a higher density of clinical-vocabulary tokens, for example, explicit references to medication, therapy modalities, and episode descriptions. These may be the easier-to-separate classes in the supervised setting.
- EDA on the full Sarkar dataset is complete. All seven labels are well-represented, with sufficient row counts per class. The Normal class has noticeably shorter average post lengths compared to other classes, while Bipolar and Personality Disorder posts tend to be longer with more clinical vocabulary.

11.2 Baseline Model

Feature engineering (NB-03) produced a TF-IDF matrix of 50,000 features (unigrams + bigrams, sublinear_tf), 10 NRC emotion lexicon scores, and 5 post-level surface statistics per sample. LDA topic modeling (NB-07) added 12 topic probability features using k=12 topics selected based on prior coherence analysis.

A leakage probe was run first: Logistic Regression trained on surface statistics only (word count, sentence count, punctuation density, capitalization ratio, average word length) achieved macro-F1 = 0.2183, well below the 0.40 threshold, confirming that no subreddit-style leakage exists in the feature set.

Model	Macro-F1	Weighted-F1	Configuration
Logistic Regression	0.7502	~0.75	C=10.0, L2 penalty, 5-fold CV
Linear SVM	0.7520	~0.75	C=1.0, 5-fold CV
XGBoost (Bayesian-tuned)	0.7965	~0.82	Optuna search, max_depth=6, lr=0.254, n_estimators=320

Both classical baselines exceed the originally expected range (0.62–0.72), indicating that the TF-IDF representation captures sufficient signal for 7-class separation. XGBoost with Bayesian

hyperparameter tuning and the full combined feature set (TF-IDF + engineered + LDA) is currently running and expected to improve over the linear baselines. DistilBERT fine-tuning (EXP-4) requires a GPU and will be run on Google Colab.

12. Conclusions

Pending complete model runs, we expect the following directional outcomes, which are working hypotheses that the final report will validate or refute against measured numbers.

On the in-distribution test set, we expect DistilBERT to outperform classical baselines (Logistic Regression, SVM, XGBoost) by approximately 5 to 12 macro-F1 points. The headroom is bounded by the relatively small training set (~4,800 posts after the 80/20 split) — published transformer gains on much larger corpora may not transfer fully at this scale.

Initial results confirm that classical NLP models achieve strong performance on the 7-class mental health classification task. Both Logistic Regression (macro-F1 = 0.7502) and Linear SVM (macro-F1 = 0.7520) exceeded the expected baseline range, indicating that TF-IDF features capture meaningful signal. The leakage probe (macro-F1 = 0.2183) confirms the models learn from lexical content rather than surface stylistic cues. We expect DistilBERT to outperform classical baselines by approximately 5 to 10 macro-F1 points based on published literature.

Per-class precision and recall, rather than aggregate macro-F1 alone, will be the operative metrics for assessing whether the models distinguish the lexically overlapping Stress, Depression, and Anxiety classes versus learning subreddit-style cues. SHAP token attributions and LDA topic clusters are expected to surface interpretable lexical and topical patterns for each class, for example, somatic-anxiety vocabulary clustering with Anxiety, manic-energy vocabulary with Bipolar Disorder, identity-disturbance vocabulary with Personality Disorder, and to surface specific tokens responsible for the most consequential routing decisions.

13. Limitations

- **Cross-source heterogeneity.** Reddit and Media Text differ in post format (full Reddit posts vs. shorter statements), label granularity, and labeling methodology. This is merged into a single training set; cross-source numbers are reported as deliberate generalization probes, not as in-distribution performance.

- **Self-report bias and label-proxy noise.** Posts are self-reported and unverified by clinical professionals. Labels are derived from subreddit membership and other online classifications rather than clinical diagnosis, introducing label noise and the risk of subreddit-style leakage. Mitigations are described in section 6, Step 1, and section 7.2.
- **Temporal generalization.** Mental-health language on social media changes over time, such as post-COVID shifts, slang turnover, and platform moderation changes. Models trained on historical data may underperform on future posts.
- **GPU access constraints.** DistilBERT fine-tuning requires a GPU. We plan to use Kaggle Notebooks (free P100/T4) as the canonical training environment with Google Colab as a development backup. Limited GPU time may constrain the number of hyperparameter search iterations we can run for EXP-4.
- **Class overlap.** Stress, Depression, and Anxiety language overlaps significantly. Aggregate macro-F1 may understate or overstate model quality if the overlap is unevenly distributed across classes; this is why per-class precision and recall are first-class outputs of every experiment.
- **Ethical scope.** This project does not deploy predictions in any clinical context. The Streamlit demo is for academic demonstration only and includes a persistent disclaimer to that effect.
- **Training-set size for transformer fine-tuning.** Training posts after the 80/20 split is enough not to be a concern for transformer fine-tuning by modern standards. Reported DistilBERT macro-F1 may need recalibration after final cross-validation results are available.

14. Future work

- Cross-platform generalization: extend evaluation to additional mental-health data to test platform-level robustness beyond the Reddit and media training distribution.
- Domain-specialized transformers: evaluate MentalBERT or mental-RoBERTa, pre-trained on mental-health text, as alternatives to general-domain DistilBERT, likely the highest-leverage extension after the present project.
- Longitudinal modeling: model temporal trajectories of individual users (post-sequence modeling with RNNs or temporal transformers) to capture clinical course rather than single-post snapshots.
- Multimodal behavioral signals: incorporate user engagement metadata like post frequency, upvote ratio, and reply depth as auxiliary features.
- Clinical validation: collaborate with mental-health professionals to validate LDA topic clusters and SHAP-identified keywords against established symptom criteria (DSM-5).
- Active learning: implement a feedback loop where flagged predictions are reviewed and corrected to improve the model over time and reduce labeling cost.

15. Contributions

Team Member	Role	Key Responsibilities
Jesigga Sigurdardottir	Data Engineer	Data acquisition (Kaggle API), preprocessing pipelines, leakage-pattern stripping, stratified 80/20 split, SMOTE setup. EDA on Media Text maintains the preprocessing report and label-mapping dictionary. Perform data preprocessing and cleaning on the data as well as apply SMOTE to the training dataset for future analysis.
Ravikumar Komandur Narayanan	NLP Feature Engineer	TF-IDF vectorization, n-gram analysis, LIWC-inspired lexicon counts, and post-level surface statistics. LDA topic modeling, coherence-based k selection, pyLDAvis visualization, and topic-by-class cross-tabulation. Word clouds and per-class lexical fingerprints.
Gunanidhi Ramakrishnan	ML Modeling Lead	Logistic Regression, Linear SVM, and XGBoost training and tuning (5-fold CV, Bayesian search). Owns utils/common.py (shared metrics, label map, random seed). Cross-source EXP-5 and OOD EXP-6 evaluation notebooks; final comparison and confusion-matrix figures.

Nathan Howland	Deep Learning & Deployment	DistilBERT fine-tuning (HuggingFace Transformers), early-stopping configuration, class-weighted training. SHAP TreeExplainer (XGBoost) and SHAP partition explainer (DistilBERT) figures, including OOD-routing attributions. Streamlit demo application; lead on paper write-up and slide deck preparation.
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16. References

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